

For the week of September 29-October 03

1) Determine, with justification, if the following integral converge or diverge.

(a) $\int_0^\infty \sin x \, dx$

(b) $\int_1^\infty \frac{dx}{x \sqrt[3]{x^2+1}}$

(c) $\int_0^\infty \frac{1}{x^2-3x+2} \, dx$

(d) $\int_1^\infty \frac{x^2+5}{\sqrt{x^5-x^3}} \, dx$

(e) $\int_1^\infty \frac{1+e^{-x}}{\sqrt{x}} \, dx$

(f) $\int_1^\infty \frac{e^{-x}}{\sqrt{x}} \, dx$

(g) $\int_{-\infty}^\infty \frac{dx}{(x^2+x+1)^2}$

(h) $\int_0^\infty \frac{x^2+1}{x^4+1} \, dx$

2) For what values of p does the following integral converge?

(a) $\int_0^\infty \frac{x^p}{\sqrt{x^2-2x+1}} \, dx$

(b) (Harder) $\int_0^1 \frac{x^p}{\sqrt{1-x^4}} \, dx$

3) Determine if the following integrals converge or diverge. If they diverge, explain why. If they converge, evaluate them.

(a) $\int_4^\infty e^{-\sqrt{x}} \, dx$

(b) $\int_{-2}^2 \frac{6x^2+8}{x^3+4x} \, dx$

(c) $\int_0^\infty \frac{1}{x^{1/2}+x^{3/2}} \, dx$

(d) $\int_2^\infty \frac{1}{\ln x} dx$

- 4) You are given the fact that $\int_0^\infty e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$. Use it to evaluate

$$\int_2^\infty e^{-\frac{(x-2)^3}{5}} \sqrt{x-2} dx.$$

- 5) In statistics, there are several functions associated with a random variable X . Two of them are its probability density function(PDF) f_X and its cumulative distribution function(CDF) F_X . For probability distributions supported on $[0, \infty)$, they are related through the following equation:

$$F_X(x) = \int_0^x f_X(t) dt.$$

One critical property of CDFs is that $\lim_{x \rightarrow \infty} F_X(x) = 1$. For each function below, determine the value of A so that it is the PDF of a random variable. That is, so that the associated CDF satisfies the above property. The parameters λ , σ , α , and β are strictly positive constants.

(a) $f_X(x) = Ae^{-\lambda x}$. (Exponential distribution with rate λ)

(b) $f_X(x) = \frac{x}{\sigma^2} e^{Ax^2}$. (Rayleigh distribution with scale σ)

(c) $f_X(x) = A \frac{(x/\alpha)^{\beta-1}}{(1+(x/\alpha)^\beta)^2}$. (Log-logistic distribution with scale α and shape β)

- 6) The expected value of a random variable X with a distribution supported on $[0, \infty)$ is given by

$$E[X] = \int_0^\infty x f_X(x) dx$$

where f_X is its PDF. For each PDF of a random variable below, calculate the expected value of X . The parameters λ , σ , and α are strictly positive constants.

(a) $f_X(x) = \frac{\alpha}{\lambda} \left(1 + \frac{x}{\lambda}\right)^{-(\alpha+1)}$. (Pareto distribution of the second kind with shape α and scale λ . Also assume that $\alpha > 1$.)

(b) $f_X(x) = \frac{\sqrt{2}}{\sigma\sqrt{\pi}} e^{-\frac{x^2}{2\sigma^2}}$. (Half-normal distribution with scale σ)

- 7) For each function below, find a value of a so that the resulting integral diverges.

(a) $\int_{-1}^1 \frac{ax^2+x}{2x^2-4x+2} dx$

(b) $\int_0^{10} \frac{1}{\sqrt{x(x^2-2ax+a^2)}} dx$

(c) $\int_1^\infty e^{-(a^2-a)x} \sin^2(ax) dx$

- 8) Determine the convergence of the following integrals.

(a) $\int_0^\infty \frac{\sin x}{x} dx.$

(b) $\int_0^\infty \left| \frac{\sin x}{x} \right| dx.$

Hint a: Use integration by parts.

Hint b: $|\sin x| \geq \sin^2 x$, also, $\sin^2 x = \frac{1 - \cos(2x)}{2}$.

- 9) The average value of a function can also be defined on an infinite interval. The average value of f on $[0, \infty)$ is the quantity

$$\lim_{x \rightarrow \infty} \frac{1}{x} \int_0^x f(t) dt.$$

Compute the average value of $\arctan x$ on $[0, \infty)$.

The following problems are for practice only and will not appear on quizzes.

I) Textbook section 7.6, problems # 6-41, 43-45, 54-56, 59-62.

II) Textbook section 7.7, problems # 1-34, 38-42.